Embodiment of metaphors with visual semantics in short and long contexts: An ERP study

Intro. According to embodied cognition, the brain's cognitive processes are influenced by and grounded in the sensorimotor experiences of the body (Foglia & Wilson, 2013). Supporting this view, some have reported modality-specific behavioral differences in the brain for visual (color only), tactile, and auditory adjectives in literal language (Platonova & Miklashevsky, 2025), whereas others reported differential electroencephalography (EEG) activations for auditory and motion/action sentences in metaphoric contexts (Schmidt-Snoek et al., 2015). The gaps in knowledge are (1) whether visual metaphoric sentences also show modality specific neural activations, and (2) the extent to which these neural activations are context dependent. In this study, we hypothesized that (1) the comprehension of visual semantics in literal and metaphoric contexts would recruit visual cortex, and (2) the degree of the visual cortex recruitment would depend on context length (short vs. long).

Methods. 31 undergraduates (ages 18-25) participated in the EEG experiment. The stimuli consisted of 252 metaphorical (MET), literal-abstract (LA), and literal-concrete (LC) sentences with short and long contexts (Table 1). Between conditions, sentential contexts were matched in syntax and length, and target words were matched in word frequency. Norming (190 subjects) were conducted to ensure that visuality and familiarity were matched between conditions, In the EEG experiment, in each trial, participant listened to a sentence and after the sentence they did a literality judgement task (literal or non-literal).

Results and Discussion. Behavioral results are summarized in Table 2. Accuracy was higher for LC than for LA, and both were higher than MET. In correct responses, MET was judged faster than LC, and both were faster than LA. The effect of context yielded similar response accuracies between both short and long sentence types, though sentences with long contexts displayed slightly slower reaction times. In EEG results at target words, there was a context difference. In short context, LA elicited the largest N400-like negativity, followed by metaphoric and then literal abstract expressions, at the occipital electrode (Oz) that is directly above the visual cortex. This pattern aligns with our predicted LC > MET > LA recruitment of visual cortex, supporting graded embodiment. A robust concreteness effect emerged at the frontal and central sites beginning at around 200 ms. For long contexts, signals at Oz were less stable, yet a reliable concreteness effect persisted frontally at Fz. Also in long contexts, MET and LA elicited a more positive late positivity (LP), centrally and parietally distributed (Cz, Pz). We suggest that this LP pattern may reflect semantic reanalysis, discourse-level updating, or late-stage embodiment processes.

Conclusion. These findings indicate that visual embodiment in both metaphoric and literal contexts is strongest when the sentential context is minimal and weakens as discourse builds, shifting processing from sensory recruitment toward higher-level integration.

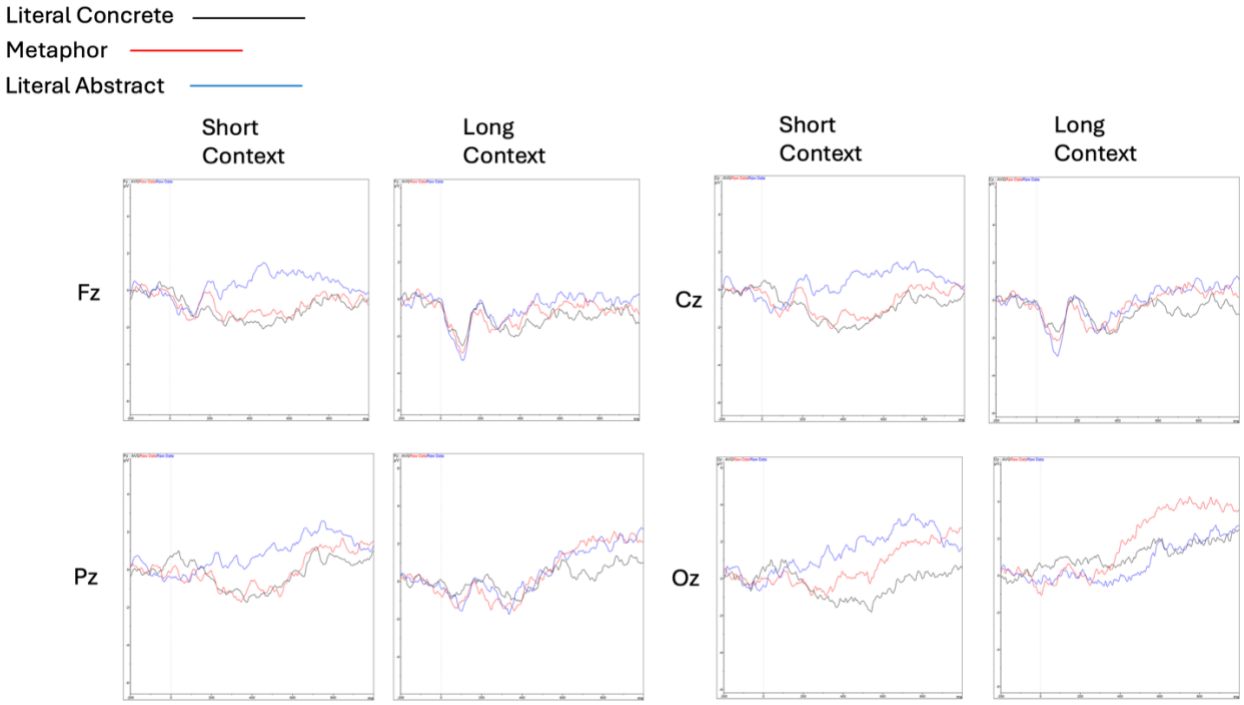
Item	Context Type	Sentence Type	Sentence	Mean Frequency (HAL)
1	short	Metaphor	This comedian's humor was quite **dark** …	8.6
1	short	Literal-concrete	This area's alleys were quite **dark** …	8.7
1	short	Literal-abstract	This comedian's humor was quite **disturbing** …	8.8
2	long	Metaphor	The discussion had by the board contained a **spectrum** …	8.6
2	long	Literal-concrete	The rainbow brought by the storm displayed a **spectrum** …	8.7
2	long	Literal-abstract	The discussion had by the board contained a **variety** …	8.8

Table 1: Example sentences in short and long contexts. Target words are bolded. "…" represent post-target context.

Sentence Type	Average Response Accuracy	Average Response Time
Metaphor	0.79	1921 ms
Literal-concrete	0.86	2001 ms
Literal-abstract	0.80	2365 ms

Table 2: Average response accuracy and response times of each condition.

Figure 1: Contrasts in amplitude elicited by each sentence type and context length at the Fz, Cz, Pz, and Oz electrodes.



References: [1] Foglia, L., & Wilson, R. A. (2013). [2] Platonova, O., & Miklashevsky, A. (2025). [3] Schmidt-Snoek, G. L., Drew, A. R., Barile, E. C., & Agauas, S. J. (2015).